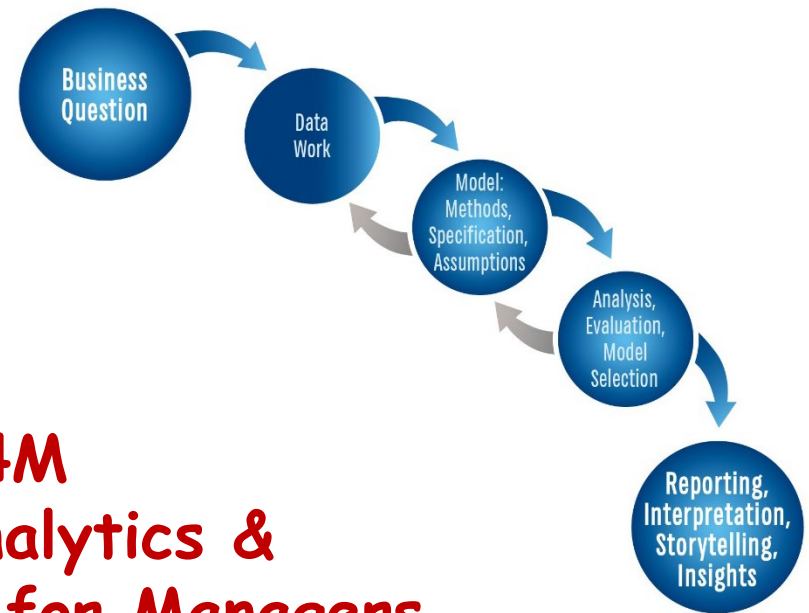




PAML4M
**Predictive Analytics &
Machine Learning for Managers**

5. Variable Selection

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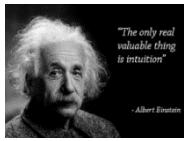
Prof. J. Alberto Espinosa

Last updated 8/1/2026





Dimensionality



Dimensionality: Intuition

- **Dimensionality** is about models having **too many variables**
- Is it good to **add variables** to a model? → **Up to a point!!**
- Is it good to **remove variables** from a model? → **Up to a point!!**
- It is a **tradeoff** → known as the **variance vs. bias** tradeoff
- **Bias** → when model coefficients don't reflect the true effects.
 - For example, the body weight of a vehicle may have a negative effect on gas mileage. But this effect may be biased if the true effect is due to engine size (which correlates with weight)
- **Variance** → If you fit the same model with different random samples of the data and you get very different results, the model suffers from high variance (i.e., it is not stable → predictions are not reliable).

What's better, more or less?

- Is it better to **add** or **remove** predictors in a model?
 - **Fewer** predictors:
 - **Good:** **Simpler** model; **Less variance**
 - **Bad:** Lower **R^2 s**; higher **MSE**; more **bias** due to omitted variables
 - **Omitted variable bias** → the **included** variables **pick up** some of the **effects** of the **omitted** variables (e.g. is engine size a predictor of car mileage? If we don't control for vehicle weight and horsepower the effect of engine size may be biased because some of its effect will be due to vehicle weight or horsepower.
 - **More** variables:
 - **Good:** Higher **R^2 s**; Lower **MSE**
 - **Less bias**; more **complex**; **more variance**
- The **Curse of Dimensionality !!** (e.g., multicollinearity)

Dimensionality Issues

Dimensionality: Bias vs. Variance

- Consider the following 3 regression models:
 - 1-Variable Model (1) $\text{mpg} = \beta_0 + \beta_{\text{Horsepower}} (\text{Horsepower}) + \varepsilon$
 - 1-Variable Model (2) $\text{mpg} = \beta_0 + \beta_{\text{Size}} (\text{Size}) + \varepsilon$
 - 2-Variable Model (3) $\text{mpg} = \beta_0 + \beta_{\text{Horsepower}} (\text{Horsepower}) + \beta_{\text{Size}} (\text{Size}) + \varepsilon$
- If Horsepower and engine Size were perfectly uncorrelated (i.e., truly independent), it can be shown mathematically that:
 - ✓ $\beta_{\text{Horsepower}}$ and β_{Size} in (1, 2 and 3) are unbiased
 - ✓ $\beta_{\text{Horsepower}}$ and β_{Size} in (3) are identical to those of (1) and (2)
 - ✓ The R^2 of (3) = R^2 of (1) + R^2 of (2) → same results
- However, if Horsepower and Size are correlated, then:
 - ✓ $\beta_{\text{Horsepower}}$ and β_{Size} in Reduced Models (1) and (2) are biased → variables included in the model pick up the effect of omitted variables → this is called “omitted variable bias”
 - ✓ $\beta_{\text{Horsepower}}$ and β_{Size} in (3) unbiased but the model has more variance

	wt	qsec	cyl	displ
wt	1.00	-0.17	0.78	0.89
qsec	-0.17	1.00	-0.59	-0.43
cyl	0.78	-0.59	1.00	-0.90
displ	0.89	-0.43	-0.90	1.00

Predictor	Est	p	Est	p	Est	p	Est	p
Intercept	37.28	<0.001	-0.11	0.624	37.68	<0.001	29.6	<0.001
wt	-5.26	<0.001					-4.52	<0.001
qsec			1.48	0.002				
cyl					-2.88	<0.001		
displ							-0.04	<0.001
R-Sq	0.75		0.58		0.72		0.72	
Adjusted R-Sq								

Dimensionality Illustration

```
cor(cbind(wt, qsec, cyl, disp))
```

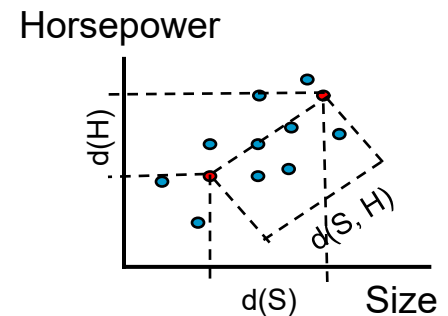
	wt	qsec	cyl	disp
wt	1.00	-0.17	0.78	0.89
qsec	-0.17	1.00	-0.59	-0.43
cyl	0.78	-0.59	1.00	0.90
disp	0.89	-0.43	0.90	1.00

```
mpg ~ wt + qsec + cyl + disp, data = mtcars
```

Predictor	Just wt		Just qsec		Just cyl		Just disp	
	Est	p	Est	p	Est	p	Est	p
Intercept	37.28	< 0.001	-5.11	0.614	37.88	< 0.001	29.6	< 0.001
wt	-5.34	< 0.001						
qsec			1.41	0.017				
cyl					-2.88	< 0.001		
disp							-0.04	< 0.001
R-Sq		0.75		0.18		0.72		0.72

Dimensionality Problems

- **Multicollinearity:** (most predominant dimensionality problem) high correlation between predictors cause the model to be unstable (e.g., dropping a few data points may yield substantially different results).
- **Overfitting:** more variables force the model to fit the data tighter, but overfitting models tend to be less accurate at predicting with new data.
- **Loss of Degrees of Freedom:** every added variable reduces the degrees of freedom of a model ($n - p - 1$).
- **Less Parsimony:** complex models difficult to interpret. Some variables will be highly significant, others not so much (keep them or not?).
- **Nuisance (or Noise) Variables:** adding variables that are not very relevant distorts its interpretability and predictive accuracy
- **High Model Variance !!** while adding more variables to a model reduces bias, the additional dimensions increase the variance of the model because the distance between points becomes larger



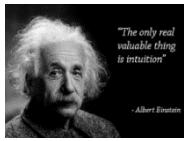
In a nutshell, **how many variables** to include in the model is a **tradeoff !!**

Addressing Dimensionality Issues

- There are a number of modeling **techniques** to deal with high dimensionality. The most popular types are:
 - ✓ **Variable Selection** (this class) – if possible, carefully **select** which variables to **include** or **exclude** and test each model
 - ✓ **Shrinkage or Regularization** (Ch.7) – when business rationale suggests that many variables need to be retained, variance caused by dimensionality can be minimized by **shrinking coefficients**, rather than removing them.
 - ✓ **Dimension Reduction Methods** (Ch.7) – correlated predictors can **combined** into fewer (i.e., reduced) **components**
 - ✓ **Non-Parametric** or **Blackbox models** (Ch.10, 11) – KNN, Trees, Neural Networks, etc. are not as affected by dimensionality.
 - ✓ **Structural Equations** (not covered) – estimation is done with two or more **related models**, rather than a single model → the outcome variable in one model is a predictor in the next model

Multicollinearity

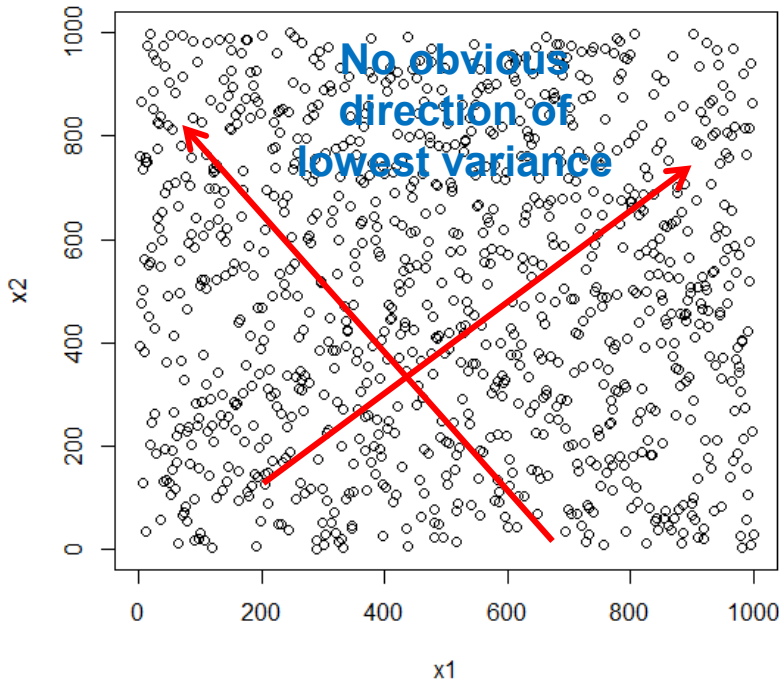
$XI(\times)$ - X's are not independent (are correlated)



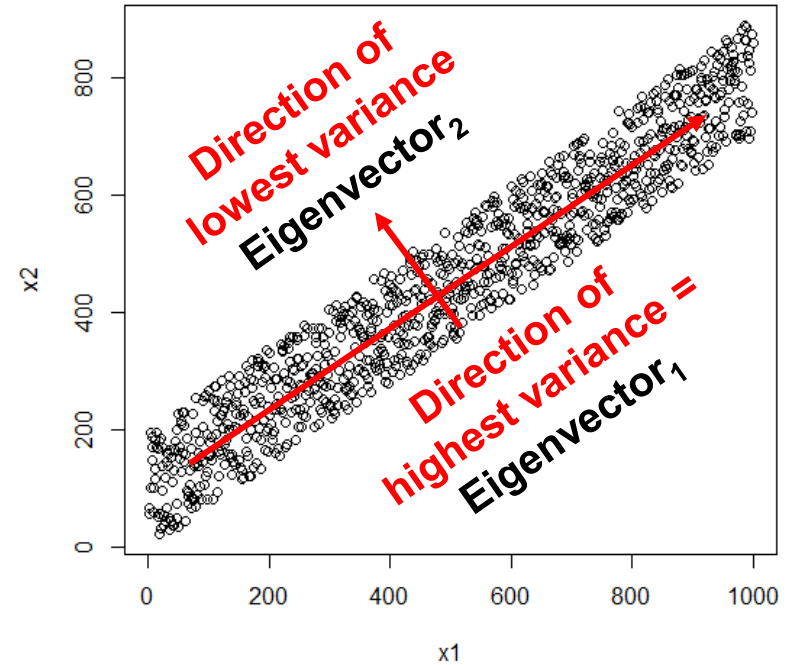
Multicollinearity: Intuition

- **Collinearity** is when **a pair** of variables have high correlation
- **Multicollinearity** is when a set of predictors are **linearly related** due to:
 - ✓ Many **correlated predictors**
 - ✓ Multivariate **linear relationships** often exist among predictors
- Multi-collinearity is **tolerable** if **moderate**; but is a **problem** if **severe**
- If some predictors are **perfectly correlated** the regression **cannot be estimated** – the solution is **indeterminate**, **standard errors** are infinite.
- If predictors are **near perfectly correlated** the model will be **unstable** (i.e., due to high variance) → the **standard errors** and **confidence intervals** will be very **large** → dropping a few observations (i.e., “**shaking the tree**”) will **change** the **results** substantially.
- Therefore, models with **severe** multicollinearity may do **deceptively well** with a random **training** sub-sample, and **poorly** with other data.

Multicollinearity, Eigenvalues & Eigenvectors



Predictors x_1 and x_2 are uncorrelated
→ Low collinearity condition



x_1 and x_2 are correlated
→ High collinearity condition

Eigenvectors → the direction of highest variance, second highest, etc.

Eigenvalues → the actual variance along each eigenvector

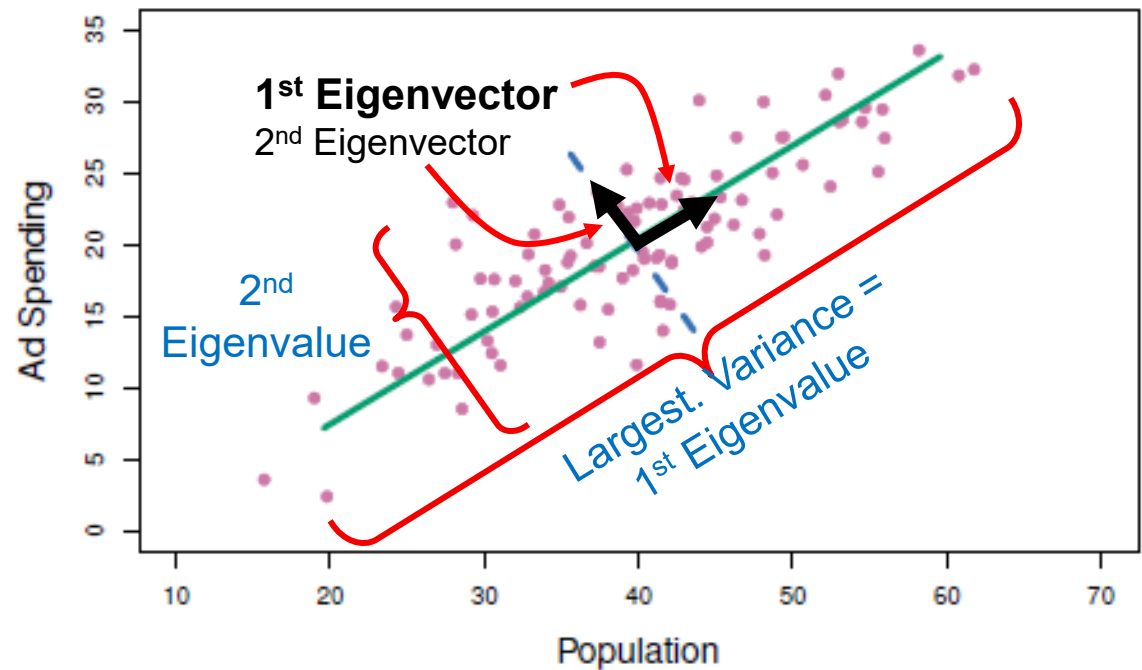
- If there are N predictors, there are N Eigenvector / Eigenvalue pairs
- Eigenvectors are ordered, from highest (1st) to lowest (n^{th})

Eigenvectors and Eigenvalues: Illustration

Let's illustrate with 2 variables like Population size and Ad Spending.

Think of PCs as axes centered at the mean and rotated into the direction of highest variance and

It is clear from this plot that Ad Spending and Population are correlated. But the resulting eigen vectors are not.



The ratio of the largest to lowest eigenvalue is an indication of how linearly related the data is → larger ratios = more multicollinearity !!

Testing for Multi-Collinearity

Testing for Multi-Collinearity

- You can test for multi-collinearity in **two ways**:
 - ✓ Evaluating the overall collinearity condition of the **entire model** → the **Condition Index (CI)** is a most popular model collinearity indicator
 - ✓ Once/if you determine that your model suffers from multi-collinearity, you can identify **which predictors** contribute the most to this collinearity condition → The **Variance Inflation Factor (VIF)** is a most popular indicator of a **predictor's contribution** to collinearity

Condition Index (CI)

- First, analyze the **correlation matrix** and inspect for:
 - **Desirable** correlations → **high** between the **outcome** and any **predictor**; and
 - **Undesirable** correlations → should be **low** among **independent** variables → if not, multicollinearity may be an issue.
- Run your regression model and extract the model's **CI**
- The **CI** is the square root of the ratio of the **largest Eigenvalue** to the **smallest Eigenvalue** of the correlation matrix.
 - ✓ **Low ratio** → **low multicollinearity** (no directional patterns in the data)
 - ✓ **High ratio** → **high multicollinearity** (clear directions of high variance)

Rule of thumb: **CI < 10 no concern with multicollinearity**

CI < 30 tolerable multi-collinearity

30 < CI < 50 some concern

CI > 50 severe multi-collinearity → must fix !!

Variance Inflation Factor (VIF)

- If the model suffers from **multicollinearity** (i.e., the **CI** is **high**) → then, the **variance** of the model, and thus **standard errors** (and **confidence intervals**) of the regression β coefficients will be **large**
- But, **how much?** It will depend on how multicollinear is the corresponding predictor with the other predictors
- The **VIF** measures the variance **inflation** contribution of **each predictor**
- The **VIF** for a given predictor X_i is obtained by fitting a regression model, using all the other independent variables to predict X_i
- If the R_i^2 of this regression is very high, it means that X_i is highly collinear with the remaining predictors
- The VIF_i of the predictor X_i is a function of this R_i^2 :

$$\checkmark \quad VIF_i = \frac{1}{1-R_i^2}$$

✓ Rule of thumb: $VIF_i < 10$ no problem → $R_i^2 < 0.90$

$VIF_i \geq 10$ too high → $R_i^2 \geq 0.90$



Multicollinearity Testing

- {library} Topic
Search keywords
- {olsrr} Condition Index (CI)
`coll <- ols_eigen_cindex(lm.fit)`
- {olsrr} Variance Inflation Factors (VIFs)
`ols_vif_tol(lm.fit)`

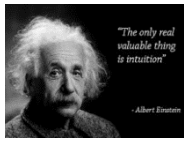
Correcting for Multi-Collinearity

Correcting for Multi-Collinearity

- If the $CI < 50 \rightarrow$ Do nothing, the model is relatively stable
- If the $CI > 50 \rightarrow$ Inspect each VIF
 - ✓ If all predictors' $VIF < 10$ (i.e., no questionable predictors)
 - ❖ A predictor is probably highly correlated with the intercept $\rightarrow$ this happens sometimes, for example if values are very large and have little variance (almost like a constant)
 - ❖ Try centering all variables (i.e., eliminate the intercept)
 - ❖ Try standardizing all variables (i.e., make the variance = 1)
 - ✓ If a predictor's $VIF > 10$, remove it if you can
 - ✓ If you can't remove a predictor $\rightarrow$ get more data
 - ✓ If you can't do either $\rightarrow$ try multicollinearity-robust models (e.g., PCR, PLS, Ridge, LASSO, etc., we will discuss these later)

Variable Selection Methods

$XI(\times)$ - X's are not independent (are correlated)



Variable Selection: Intuition

- As we discussed earlier, two **key** aspects of predictive modeling are:
(1) Model Selection and **(2) Model Specification**
- We also discussed that two key aspects of **Model Specification** are:
(1) Data Pre-Processing (e.g., transformations, etc.) and
(2) Variable Selection
- In this section, we focus on **Variable Selection**, i.e., the **selection** of the most **appropriate predictors** for the model
- Predictors can be selected based on:
 1. **Business knowledge** – always **best** to select predictors that make business sense for the particular problem at hand.
 2. **Statistical fit** – but it is also important to evaluate **statistically** the various **candidate sets** of predictors
- The concept of **Dimensionality** is at the heart of **Variable Selection**

Selecting Predictors

- **Variable selection** → which variables to retain in or remove from the model
- It is **safe** to **remove** variables that are **not statistically significant**
- But it is **not OK** to remove **significant** variables, unless you have sound justification or serious dimensionality issues → doing so will **bias** your model

Variable Selection Tradeoffs

- **Smaller** models are more **parsimonious** (i.e., simpler to interpret) and have **less variance**
- But having fewer predictors in the model may present an **incomplete picture**
- If predictors have **little correlation**, it is not a big problem to omit some variables for simplicity – the results will be similar with a few small models, than with a larger model with all predictors.
- But if some predictors have **high correlation**, omitting one may **bias** the coefficient of the variables retained
- **Larger** models are **less biased**, but they have **more variance**
- The **general rule** is to select the **largest** model with **tolerable multi-collinearity**

Comparing Models

Nested Models with Different Specifications

- When one model specification is a **subset** of a larger model specification you can test if the larger model has significantly **higher predictive power**

Different Models

- To **compare** two models that are very **different** you need to evaluate them using one of the **cross-validation** methods (covered **later**) → fit the models with a (**train**) subset and test/compare the models' predictive power with a different (**test**) subset

Comparing Nested Models

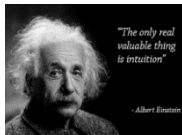
Comparing Nested Models

- The **larger** model **ALWAYS** has **more predictive power** than the smaller model. Key question: is the **difference significant?** i.e., is it worth the added complexity of a larger model?
- You can conduct an **ANOVA F-Test** to evaluate if a larger model has statistically stronger **predictive power** (e.g., higher **R^2** , lower **MSE**)
- If the **F-Test** shows **no significant improvement** with the additional variables, go with the **reduced model** because it is simpler
- If the **F-Test** is **significant** then select the **full model**
- And, of course, you can **ALWAYS** use **cross-validation** methods to compare the two models

Variable Selection Methods

- **Subset Comparison** – if you have 2 or a few candidate models to compare, you can test them against each other with ANOVA tests
- **Step Methods** – if you have a large and a small subsets of predictors, and would like to test various combinations between them, you can progressively add (Forward) or remove (Backwards) variables, or both back and forth (Stepwise)

Subset Comparison



Subset Comparison: Intuition

You can test any 2 **nested** models (one is a **subset** of the other):

Large vs. **Small** (or Restricted):

Small Model: $Y = \beta_0 + \underbrace{\beta_1(X_1) + \beta_2(X_2)}_{\text{Nested}} + \varepsilon$

Large Model: $Y = \beta_0 + \underbrace{\beta_1(X_1) + \beta_2(X_2)}_{\text{Nested}} + \dots + \beta_n(X_n) + \varepsilon$

- The **Large** model's **MSE** < **Smaller** model's **MSE**
- But is the difference statistically **significant**? → **Test it !!**
- Use the **ANOVA F-Test** to compare the models
 - If **significant**, select the **Large Model**
 - retain the **Small Model otherwise**
- **Fun fact**: if any of the **added coefficients** to the Full Model are **significant**, the **ANOVA F-Test** will also be **significant**.

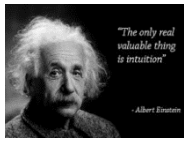


Subset Selection

- {library} Topic
Search keywords
- Subset Comparisons
`anova(lm.reduced, lm.large, lm.full)`



Step Methods



Step Methods: Intuition

Suppose you have P possible predictors $\rightarrow$ 2 extreme models:

Small Model (NO or few predictors): $Y = \beta_0 + \beta_1(X_1) + \varepsilon$

Large Model (Many predictors): $Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \dots + \beta_P(X_P) + \varepsilon$

- You may want to ANOVA **test all possible models** between the **Small** and **Large** models
- But if P is **large**, the number of possible models will grow exponentially ($2^P - 1$) and it may not be computationally practical to test all of them.
 - 10 variables $\rightarrow 2^{10} = 1,024$ models
 - 20 variables $\rightarrow 2^{20} = 1,048,576$ models
- Step methods resolve this by **adding/removing 1 variable at a time**
- You can start with the **Small** model and **add** variables (**forward**) towards the **Large** model; or start with the **Large** model and **remove** variables (**backwards**) towards the **Small** model

Step Methods

These are **NOT modeling methods** → all resulting models are OLS or GLM. Step are **process method** that add and/or remove variables to **identify** an optimal **model specification**.

Progressively **adding/removing** variables and **testing**:

- ✓ **Forward step selection**: starts with **0 (or a small set of) predictors**; then **add** the most significant predictor; then the next one; continue until **none** of the **remaining** variables are **significant** when added to the model
- ✓ **Backward step elimination**: starts with **all (or a large set of) predictors**; then **remove** the least significant variable; then the next; and so on until **no more remaining** variables are **significant** when added to the model
- ✓ **Stepwise method**: similar to forward or backwards, except that variables can be **added** or **removed** in any iteration. The **p-value** for inclusion or exclusion can be set separately

Forward, Backward and Stepwise

Forward Step Selection



- **Start** with the **Null** (or **reduced**) model; then try all **single-variable** models; select the **most significant variable**.
- Try each of the remaining variables, 1 at time; select the **most significant** variable, and so on.
- **Stop** when **no** more variables are **significant** when added to the model

Predictors

	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X ₁₀	X ₁₁	X ₁₂
0 (Null)												
1 (Add)												
2 (Add)												
3 (Add)												
4 (Add)												
5 (Add)												
6 (Stop)												

Problem: this method **starts** with the **most biased** model → as you add variables, some of the variables in the model may lose significance

Backwards Step Elimination

- Start with the **Full** (or **large**) model; then remove the **least significant variable**; then the **next least significant**, and so on.
- Stop when **no** more of the variables left out become **significant** when there are **no more non-significant** variables to remove.

Predictors

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}
12 (Full)												
11 (Rem)												
10 (Rem)												
9 (Rem)												
8 (Rem)												
7 (Rem)												
6 (Stop)												

Problem: this method **starts** with the most **over-identified** model → may suffer from **dimensionality** issues → as you remove variables, some that were not significant may become **significant if added** again

Stepwise

- Works like **Backward** or **Forward** selection → in each step you either **remove non-significant** variables or **add significant** variables
- **Stop** when **no** more variables become significant if added to (removed from) the model and there are no non-significant variables to remove

Predictors

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}
12 (Full)												
11 (Rem)												
10 (Rem)												
9 (Rem)												
8 (Add)												
7 (Rem)												
6 (Stop)												

Significant

Problem: similar to Forward and Backwards methods, but it is **often preferred** due to the flexibility. One can **vary** the **p-value** for **inclusion** or **removal** of variables, thus giving more control of the selection process



Stepwise Selection

- {library} Topic
Search keywords
- Step Methods → Lower and upper bound models
`Hitters.null <- lm(Salary ~ 1, data = Hitters)`
`Hitters.full <- lm(Salary ~., data = Hitters)`
- Step Methods → Setting $k = 0.05$ parameter
`qchisq(0.05, 1, lower.tail = F)`
- Stepwise Variable Selection
`Hitters.step <- step(Hitters.full,`